321. The number of unpaired electrons in Zn^{2+} is

(a) 2

(b) 3

(c) 4

(d) 0

Answer: D — (d) 0

Chemical reason: The correct choice is 0. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

322. The trace metal present in insulin is

(a) Iron

(b) Cobalt

(c) Zinc

(d) Manganese

Answer: C — (c) Zinc

Chemical reason: Zinc is associated with insulin storage: insulin in pancreatic β-cell granules is stabilized in zinc-containing hexameric form. Hence Zn is the trace metal linked with insulin.

323. The chemical name of borax is

(a) Sodium orthoborate

(b) Sodium metaborate

(c) Sodium tetraborate

(d) Sodium tetraborate decahydrate

Answer: D — (d) Sodium tetraborate decahydrate

Chemical reason: Borax is sodium tetraborate decahydrate, $Na_2B_4O_7 \cdot 10H_2O$. "Borax" normally refers to this hydrated crystalline salt.

324. Hydrogen is not obtained when zinc reacts with

(a) Cold water

(b) Dilute H_2SO_4

(c) Dilute HCl

(d) Hot 20% NaOH

Answer: A — (a) Cold water

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives Cold water. The choice is consistent with the expected oxidation state and stability of the product.

325. The metal which gives hydrogen on treatment with acid as well as sodium hydroxide is

(a) Iron

(b) Zinc

(c) Copper

(d) None of the above

Answer: B — (b) Zinc

Chemical reason: Zinc reacts with acids to release H_2 and also dissolves in strong NaOH to form zincate with H_2 evolution. This dual reaction reflects the amphoteric nature of zinc and $ZnO/Zn(OH)_2$.

326. To protect iron against corrosion, the most durable metal plating on it, is

(a) Nickel plating
(b) Tin plating
(c) Copper plating
(d) Zinc plating

Answer: D — (d) Zinc plating

Chemical reason: Zinc plating is especially durable for protecting iron because zinc is more reactive than iron and continues to protect sacrificially even if the coating is scratched.

327. The compound $ZnFe_2O_4$is

(a) A normal spinel compound
(b) Interstitial compound
(c) Covalent compound
(d) Co-ordination compound

Answer: A — (a) A normal spinel compound

Chemical reason: The correct choice is A normal spinel compound. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

328. ZnO when heated with BaO at 1100°C gives a compound. Identify the compound

(a) $BaZnO_2$
(b) BaO_2+Zn
(c) $BaCdO_2$
(d) Ba+ZnO_2

Answer: A — (a) $BaZnO_2$

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives $BaZnO_2$. The choice is consistent with the expected oxidation state and stability of the product.

329. Zn gives hydrogen gas with $H_2SO_4$and HClbut not with HNO_3 because

(a) $NO_2$is reduced in preference to H_3O^+
(b) $HNO_3$is weaker acid than $H_2SO_4$and HCl
(c) Znacts as oxidising agent when reacts with HNO_3
(d) In electrochemical series Znis placed above the hydrogen

Answer: B — (b) $HNO_3$is weaker acid than $H_2SO_4$and HCl

Chemical reason: Nitric acid is an oxidizing acid. With zinc it is reduced to nitrogen oxides or other nitrogen products rather than H^+ simply being reduced to H_2, so hydrogen gas is generally not evolved.

330. The metal used for making radiation shield is

(a) Aluminium

(b) Iron

(c) Zinc

(d) Lead

Answer: D — (d) Lead

Chemical reason: Lead is used for radiation shielding because its very high density and high atomic number give strong attenuation of X-rays and gamma rays.

331. Which of the following metal is obtained by leaching out process using a solution of NaCN and then precipitating the metal by addition of zinc dust

(a) Copper

(b) Silver

(c) Nickel

(d) Iron

Answer: B — (b) Silver

Chemical reason: Silver is extracted by cyanide leaching: Ag forms the soluble dicyanoargentate complex and is then displaced from solution by zinc, which is more electropositive.

332. While extracting an element from its ore, the ore is ground and leached with dilute KCN solution to form the soluble product potassium argentocyanide. The element is

(a) Lead

(b) Chromium

(c) Manganese

(d) Silver

Answer: D — (d) Silver

Chemical reason: Formation of soluble potassium/sodium argentocyanide is characteristic of silver cyanidation. The silver complex is later reduced/displaced to recover metallic Ag.

333. In Mc Arthur Forest method, silver is extracted from the solution of $Na[Ag(CN)_2]$ by the use of:

(b) Fe

(b) Mg

(c) Cu

(d) Zn

Answer: D — (d) Zn

Chemical reason: In the MacArthur–Forrest cyanide process, zinc displaces silver from its soluble cyanide complex because Zn is more electropositive than Ag.

334. Iron obtained from blast furnace is known as

(a) Wrought iron
(b) Cast iron
(c) Pig iron
(d) Steel

Answer: C — (c) Pig iron

Chemical reason: The molten iron tapped directly from a blast furnace is pig iron. It contains relatively high carbon plus silicon, phosphorus, sulfur, and manganese impurities.

335. Extraction of silver from commercial lead is possible by

(a) Mond's process
(b) Park's process
(c) Haber's process
(d) Clark's process

Answer: B — (b) Park's process

Chemical reason: Parkes' process removes silver from argentiferous lead by adding zinc; silver preferentially dissolves in the zinc-rich phase, which can be separated from the molten lead.

336. Impurities of lead in silver are removed by

(a) Park process
(b) Solvey process
(c) Cyanide process
(d) Amalgamation process

Answer: A — (a) Park process

Chemical reason: Silver chemistry is governed strongly by low-solubility halides and complex formation with ligands such as NH_3 or $S_2O_3^{2-}$. Under the stated conditions this leads to Park process.

337. Park's process is used in the extraction of

(a) Iron
(b) Zinc
(c) Silver
(d) Lead

Answer: C — (c) Silver

Chemical reason: Parkes' process is specifically used for desilverization of lead—recovery of silver from lead containing small amounts of Ag.

338. From argentite (Ag_2S) ore the method used in obtaining metallic silver is

(a) Fused mixture of Ag_2S and KCl is electrolysed
(b) Ag_2S is reduced with CO
(c) $Ag_2S$is roasted to Ag_2Owhich is reduced with carbon
(d) Treating argentite with NaCN solution followed by metal displacement with zinc

Answer: D — (d) Treating argentite with NaCN solution followed by metal displacement with zinc

Chemical reason: Argentite is leached with cyanide to form a soluble silver-cyanide complex, after which zinc displaces silver from the solution. This is the cyanide/MacArthur–Forrest route.

339. In the extraction of zinc which gas is burnt in the jackets surrounding the retorts

(a) Water gas

(b) Producer gas

(c) Oil gas

(d) Coal gas

Answer: B — (b) Producer gas

Chemical reason: The correct choice is Producer gas. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

340. MacArther process is used for

(a) Hg

(b) Fe

(c) Cl

(d) O_2

Answer: A — (a) Hg

Chemical reason: The application follows from the characteristic chemical property of the material—such as catalytic activity, stability, conductivity, or corrosion resistance. That property is specifically associated with Hg.